

10/2/2026

Last time:

1. Units in  $\mathbb{Z}[\sqrt{d}]$  when  $d < 0$ .

①  $d = -1$ :  $\pm 1, \pm i$

②  $d < -1$ :  $\pm 1$

Ex. If  $d = -m^2 < -1$ , then  $\mathbb{Z}[\sqrt{d}] = \mathbb{Z}[mi]$  is

not a UFD:

$$m^2 = m \cdot m = \overset{\text{unit}}{\downarrow} (-1) mi \cdot mi.$$

Claim:  $mi$  is prime in  $\mathbb{Z}[mi]$ .

Pf. Let  $\alpha = x + miy \mid mi$ . Then

$$N(\alpha) = x^2 + m^2 y^2 \mid N(m_i) = m^2.$$

So  $m^2 \geq x^2 + m^2 y^2$ . Hence  $y = 0, \pm 1$ .

(1)  $y = \pm 1$  :  $x = 0$ . So  $\alpha = \pm m_i$  is an associate of  $m_i$

(2)  $y = 0$ :  $\alpha = x \mid m_i$ , so  $\beta = \frac{m_i}{x} \in \mathbb{Z}[m_i]$

① If  $x = \pm 1$ , then  $\alpha = \pm 1$  is a unit.

② If  $x \neq \pm 1$ , then  $\beta = \frac{1}{x} \cdot m_i \notin \mathbb{Z}[m_i]$ ,

a contradiction. (In fact, one can deduce  $x \mid m$

from  $N(\alpha) = x^2/m^2$ , but we can certainly avoid it)

So  $m_i$  is prime.

Return to

$$m^2 = m \cdot m = (-1) m_i \cdot m_i.$$

unit  
↑

↓ ↓  
prime

Since  $\frac{m}{m_i} = \frac{1}{i} = -i \notin \mathbb{Z}[m_i]$ ,  $m_i \nmid m$ . Hence,

$m_i$  will never show up in the factorization

$$m \cdot m = (\underbrace{\pi_1 \pi_2 \dots \pi_k}_m) (\underbrace{\pi_1 \pi_2 \dots \pi_k}_m).$$

## 2. Primes in $\mathbb{Z}[i]$ .

①  $\pm 1, \pm i$  are prime and they are prime factors of

2.

②  $p \equiv 3 \pmod{4}$  and its associates are prime:

$$x^2 + y^2 \neq p$$

③  $p \equiv 1 \pmod{4}$  is not prime in  $\mathbb{Z}[i]$ !

$$\exists x, y \text{ s.t. } x^2 + y^2 = p$$
$$\underbrace{\hspace{1.5cm}}_{(x+yi)(x-yi)}$$

Lem. Every prime in  $\mathbb{Z}[i]$  divides exactly one prime number.

Existence :  $\pi \mid N(\pi) = p_1 p_2 \cdots p_k \xRightarrow{\text{Euclid}} \pi \mid p_j \text{ for some } j.$

Uniqueness :  $\pi \mid p, \pi \mid q \quad (p \neq q)$

$\xRightarrow{\text{Bézout}} \exists x, y \in \mathbb{Z} \text{ s.t. } px + qy = 1.$

$\Rightarrow \pi \mid 1$

$\Rightarrow \pi \text{ is a unit } \times$

Note : The uniqueness actually proves that if  $a, b \in \mathbb{Z}$

with  $\gcd(a, b) = 1$  in  $\mathbb{Z}$ , then  $\gcd(a, b) = 1$  in  $\mathbb{Z}[i]$ .

The converse is also true.

Thm (Primes in  $\mathbb{Z}[i]$ ). Let  $\pi \in \mathbb{Z}[i]$  be prime.

1.  $\pi = \pm 1 \pm i \mid 2$ .

2.  $\pi = p \equiv 3 \pmod{4}$  or one of its associates

3.  $\pi = x + iy \mid p$  with  $x, y \neq 0$ , where  $p \equiv 1 \pmod{4}$ .

In fact,

$$N(\pi) = \pi \bar{\pi} = x^2 + y^2 = p.$$

If  $\pi$  is such a prime divisor of  $p$ , there are

8 in total:

$$\pm \pi, \pm i\pi, \pm \bar{\pi}, \pm i\bar{\pi},$$

namely, associates of  $\pi$  and  $\bar{\pi}$ . So there are exactly

8 ways of writing  $p$  as a sum of two squares.